

①

Solved questions

Date: / /

Chapter 1.1

Example 01

$$S = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$$

Y_1 = number of customers choose counter 1

Y_2 = number of customers choose counter 2

$$P(Y_1, Y_2) = ?$$

		Y_1			
		0	1	2	
Y_2	0	1/9	2/9	1/9	4/9
	1	2/9	2/9	0	4/9
	2	1/9	0	0	1/9
		4/9	4/9	1/9	1

$$\therefore P(Y_1, Y_2) = \begin{cases} 1/9 & ; \text{ if } (Y_1, Y_2) \in \{(0,0), (2,0), (0,1)\} \\ 2/9 & ; \text{ if } (Y_1, Y_2) \in \{(1,0), (0,1), (1,1)\} \\ 0 & ; \text{ elsewhere} \end{cases}$$

joint probability function

Example 02

Y_1 - number of dots occurred in die 1
 Y_2 - number of dots occurred in die 2

$$\left. \begin{matrix} 1 \leq Y_1 \leq 6, 1 \leq Y_2 \leq 6 \end{matrix} \right\}$$

every pair (Y_1, Y_2) has the same joint probability $1/36$ because, this experiment has 36 sample events and all of them are equiprobable

(2)

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so; joint distribution functions,

$$F(-1, 2) = P(\underbrace{-\infty \leq y_1 \leq -1}_{\text{because, this part has no meaning regarding this experiment}}, 0 \leq y_2 \leq 2)$$

$$= 0$$

because, this part has no meaning regarding this experiment

$$F(1.5, 2) = P(1 \leq y_1 \leq 1.5, 1 \leq y_2 \leq 2)$$

$$= \cancel{P(0, 1) + P(0, 2) + P(1, 1)}$$

$$= P(1, 1) + P(1, 2)$$

$$= \frac{1}{36} + \frac{1}{36}$$

$$= \frac{2}{36}$$

$$F(5, 7) = P(1 \leq y_1 \leq 5, 1 \leq y_2 \leq 7)$$

$$= \frac{30}{36} = \frac{5}{6}$$

Example 03

joint density function $f(y_1, y_2) = \begin{cases} 1 & ; 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1 \\ 0 & ; \text{elsewhere} \end{cases}$

$$\begin{aligned} \text{b.) } F(0.2, 0.4) &= \int_0^{0.2} \int_0^{0.4} f(y_1, y_2) dy_2 dy_1 \\ &= \int_0^{0.2} \int_0^{0.4} 1 dy_2 dy_1 \\ &= \int_0^{0.2} [y_2]_0^{0.4} dy_1 \\ &= \int_0^{0.2} (0.4 - 0) dy_1 = \int_0^{0.2} (0.4) dy_1 = 0.4 \int_0^{0.2} 1 dy_1 \\ &= 0.4 [y_1]_0^{0.2} = 0.4 (0.2 - 0) \\ &= 0.08 \end{aligned}$$

$$\begin{aligned}
 c) P(0.1 \leq Y_1 \leq 0.3, 0 \leq Y_2 \leq 0.5) &= \int_{0.1}^{0.3} \int_0^{0.5} f(y_1, y_2) dy_2 dy_1 \\
 &= \int_{0.1}^{0.3} \int_0^{0.5} (1) dy_2 dy_1 \\
 &= 0.5 (0.3 - 0.1) \\
 &= \underline{\underline{0.1}}
 \end{aligned}$$

Q1.

$$S = \{(A, B), (A, C), (B, C), (B, A), (C, A), (C, B), (A, A), (B, B), (C, C)\}$$

Y_1 - number of contracts assigned firm A

Y_2 - number of contracts assigned firm B

		Y_1			
		0	1	2	
Y_2	0	1/9	2/9	1/9	4/9
	1	2/9	2/9	0	4/9
	2	1/9	0	0	1/9
		4/9	4/9	1/9	1

$$a) p(y_1, y_2) = \begin{cases} 1/9; & \text{if } (y_1, y_2) \in \{(0,0), (0,2), (2,0)\} \\ 2/9; & \text{if } (y_1, y_2) \in \{(1,0), (0,1), (1,1)\} \\ 0; & \text{elsewhere} \end{cases}$$

$$b) F(1,0) = \underline{\underline{3/9}}$$

Q2.

	0	0	0	
x	x	x	x	→ -1
✓	x	x	x	→ 1
x	✓	x	x	→ 2
x	x	✓	✓	→ 3
✓	✓	x	x	→ 1
✓	x	✓	✓	→ 1
x	✓	✓	✓	→ 2
✓	✓	✓	✓	→ 1

		Y_1			
		0	1	2	3
Y_2	-1	$1/8$	0	0	0
	1	0	$1/8$	$2/8$	$1/8$
	2	0	$1/8$	$1/8$	0
	3	0	$1/8$	0	0

$$a) P(Y_1, Y_2) = \begin{cases} 1/8 & \text{if } (Y_1, Y_2) \in \{(0, -1), (1, 1), (1, 2), (1, 3), (2, 2), (3, 1)\} \\ 2/8 & \text{if } (Y_1, Y_2) \in \{(2, 1)\} \\ 0 & \text{elsewhere} \end{cases}$$

$$b) F(2, 1) = P(0 \leq Y_1 \leq 2, -1 \leq Y_2 \leq 1)$$

$$= P(0, -1) + P(0, 1) + P(1, -1) + P(1, 1) + P(2, -1) + P(2, 1)$$

$$= 1/8 + 0 + 0 + 1/8 + 0 + 2/8$$

$$= 4/8 = 1/2$$

Chapter 1.2

Example 04

$$S = \{R_1R_2, R_1R_3, R_2R_3, R_1D_1, R_1D_2, R_1I, R_2D_1, R_2D_2, R_2I, R_3D_1, R_3D_2, R_3I, D_1D_2, D_1I, D_2I\}$$

Y_1 - number of Republicans

Y_2 - number of Democrats

		Y_1			
		0	1	2	
Y_2	0	0	$3/15$	$9/15$	$6/15$
	1	$2/15$	$6/15$	0	$8/15$
	2	$1/15$	0	0	$1/15$
		$3/15$	$9/15$	$3/15$	1

} marginal probability function of Y_2

marginal probability function of Y_1

$$p(y_1, y_2) = \begin{cases} 1/15 & ; \text{ if } (y_1, y_2) \in \{(0, 2)\} \\ 2/15 & ; \text{ if } (y_1, y_2) \in \{(0, 1)\} \\ 3/15 & ; \text{ if } (y_1, y_2) \in \{(1, 0), (2, 0)\} \\ 6/15 & ; \text{ if } (y_1, y_2) \in \{(1, 1)\} \\ 0 & ; \text{ elsewhere} \end{cases}$$

$$p_1(y_1) = \sum_{\text{all } y_2} p(y_1, y_2) = p(y_1 = y) = \begin{cases} 3/15 & ; y = 0 \\ 9/15 & ; y = 1 \\ 3/15 & ; y = 2 \\ 0 & ; \text{ otherwise.} \end{cases}$$

marginal probability function of y_1

Example 05

$$f(y_1, y_2) = \begin{cases} 2y_1 & ; 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1 \\ 0 & ; \text{ elsewhere} \end{cases}$$

$$\left. \begin{aligned} f_1(y_1) &= \int_0^1 f(y_1, y_2) dy_2 \\ &= \int_0^1 2y_1 dy_2 \\ &= 2y_1 [y_2]_0^1 \\ &= 2y_1 \end{aligned} \right\} \begin{aligned} f_2(y_2) &= \int_0^1 f(y_1, y_2) dy_1 \\ &= \int_0^1 2y_1 dy_1 \\ &= 2 \left[\frac{y_1^2}{2} \right]_0^1 \\ &= 2 \times \frac{1}{2} = 1 \end{aligned}$$

Example 06

$$\cancel{P(Y_2=1 | Y_1)} \quad P(Y_1=0 | Y_2=1) = \frac{p(0,1)}{p_2(1)} = \frac{2/15}{8/15} = \frac{1}{4}$$

Conditional probabilities

$$P(Y_1=1 | Y_2=1) = \frac{p(1,1)}{p_2(1)}$$

$$= \frac{6/15}{8/15} = \frac{3}{4}$$

* we can say about this
"Y₁ given that Y₂=1"

$$\begin{aligned}
 P(Y_1=2 | Y_2=1) &= \frac{P(2,1)}{P_2(1)} \\
 &= \frac{0}{8/15} = \underline{\underline{0}}
 \end{aligned}$$

Example 07

$$f(y_1, y_2) = \begin{cases} 1/2 & ; 0 \leq y_1 \leq y_2 \leq 2 \\ 0 & ; \text{elsewhere} \end{cases}$$

$$f(y_1 | y_2) = \frac{f(y_1, y_2)}{f_2(y_2)} \quad \text{--- (1)}$$

Then,

$$\begin{aligned}
 f_2(y_2) &= \int_0^{y_2} f(y_1, y_2) dy_1 \\
 &= \int_0^{y_2} (1/2) dy_1
 \end{aligned}$$

$$= \frac{1}{2} (y_2 - 0) = \underline{\underline{\frac{1}{2} y_2}}$$

from (1);

$$f(y_1 | y_2) = \frac{1/2}{1/2 y_2} = \underline{\underline{\frac{1}{y_2}}}$$

$$f(Y_1 < 0.5 | Y_2 = 1.5) = \frac{f(0.5, 1.5)}{f_2(1.5)} \quad \text{--- (2)}$$

$$f(y_1 | 1.5) = \frac{1}{1.5} = \frac{2}{3}$$

$\therefore$ from (2);

$$f(Y_1 < 0.5 | Y_2 = 1.5) = \int_0^{0.5} \frac{2}{3} dy_1$$

$$= \frac{2}{3} \times \frac{1}{2}$$

$$= \underline{\underline{\frac{1}{3}}}$$

Chapter 1.3

Example of _____.

 Y_1 - number of dots appearing on die 1 Y_2 - number of dots appearing on die 2If Y_1 and Y_2 are independent;

$$P(Y_1, Y_2) = P_1(Y_1) \times P_2(Y_2)$$

$P(Y_1, Y_2) = 1/36$ ^①; for any pair (Y_1, Y_2) , the joint probability function has the same probability value of $1/36$

$$P_1(Y_1) = 1/6, \quad P_2(Y_2) = 1/6$$

$$\therefore P_1(Y_1) \times P_2(Y_2) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36} \quad \text{--- ②}$$

$$\text{①} = \text{②};$$

$$\therefore P(Y_1, Y_2) = P_1(Y_1) P_2(Y_2)$$

Therefore; Y_1 and Y_2 are independent

Example 09

 ~~$P_1(Y_1)$~~ we can test this for $(Y_1=1, Y_2=1)$ pair.

$$P(1, 1) = 6/15 = 0.4 \quad \text{--- ①}$$

$$P_1(1) = 9/15 = 0.6$$

$$P_2(1) = 8/15 \approx 0.533$$

$$P_1(1) \times P_2(1) = 0.6 \times 0.533$$

$$= 0.3198 \neq \underline{\underline{0.4}}$$

So; Y_1 and Y_2 are not independent

Example 10

$$f(y_1, y_2) = \begin{cases} 6y_1 y_2^2; & 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1 \\ 0; & \text{elsewhere} \end{cases}$$

If independent;

$$f(y_1, y_2) = f_1(y_1) \times f_2(y_2)$$

$$\begin{aligned} f_1(y_1) &= \int_0^1 f(y_1, y_2) dy_2 \\ &= \int_0^1 6y_1 y_2^2 dy_2 \\ &= 6y_1 \left[\frac{y_2^3}{3} \right]_0^1 = \underline{\underline{2y_1}} \end{aligned}$$

$$\begin{aligned} f_2(y_2) &= \int_0^1 f(y_1, y_2) dy_1 \\ &= \int_0^1 6y_1 y_2^2 dy_1 \\ &= 6y_2^2 \left[\frac{y_1^2}{2} \right]_0^1 = \underline{\underline{3y_2^2}} \end{aligned}$$

$$\begin{aligned} \therefore f_1(y_1) \times f_2(y_2) &= 2y_1 \times 3y_2^2 \\ &= \underline{\underline{6y_1 y_2^2}} \end{aligned}$$

$$\text{Also; } f(y_1, y_2) = \underline{\underline{6y_1 y_2^2}}$$

$$\therefore f(y_1, y_2) = f_1(y_1) \times f_2(y_2)$$

Therefore; y_1 and y_2 are independent

Example 14

$$f(y_1, y_2) = \begin{cases} 2y_1; & 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1 \\ 0; & \text{elsewhere} \end{cases}$$

$$\begin{aligned} E(Y_1 Y_2) &= \int_0^1 \int_0^1 y_1 y_2 f(y_1, y_2) dy_1 dy_2 \\ &= \int_0^1 2y_2 \left[\frac{y_1^2}{2} \right]_0^1 dy_2 \end{aligned}$$

$$\begin{aligned}
 &= \int_0^1 \frac{2}{3} y_2 \, dy_2 \\
 &= \frac{2}{3} \left[\frac{y_2^2}{2} \right]_0^1 \\
 &= \frac{2}{3} \times \frac{1}{2} \\
 &= \underline{\underline{\frac{1}{3}}}
 \end{aligned}$$

$$\begin{aligned}
 E(Y_1) &= \int_0^1 \int_0^1 y_1 (2y_1) \, dy_1 \, dy_2 \\
 &= \int_0^1 2 \left[\frac{y_1^2}{2} \right]_0^1 \, dy_2 \\
 &= \underline{\underline{\frac{2}{3}}}
 \end{aligned}$$

$$\begin{aligned}
 E(Y_2) &= \int_0^1 \int_0^1 y_2 (2y_1) \, dy_1 \, dy_2 \\
 &= \int_0^1 2y_2 \left[\frac{y_1^2}{2} \right]_0^1 \, dy_2 \\
 &= \underline{\underline{\frac{1}{2}}}
 \end{aligned}$$

$$\begin{aligned}
 V(Y_1) &= E[Y_1^2] - (E[Y_1])^2 \\
 &= \frac{1}{2} - \left(\frac{2}{3}\right)^2 \\
 &= \underline{\underline{\frac{1}{18}}}
 \end{aligned}$$

$$\begin{aligned}
 E[Y_1^2] &= \int_0^1 \int_0^1 y_1^2 (2y_1) \, dy_1 \, dy_2 \\
 &= \int_0^1 2 \left[\frac{y_1^4}{4} \right]_0^1 \, dy_2 \\
 &= \int_0^1 \frac{1}{2} \, dy_2 \\
 &= \underline{\underline{\frac{1}{2}}}
 \end{aligned}$$

Example 17

$$f(y_1, y_2) = \begin{cases} 3y_1 & ; 0 \leq y_2 \leq y_1 \leq 1 \\ 0 & ; \text{elsewhere} \end{cases}$$

$$\begin{aligned}
 E(Y_1 - Y_2) &= E(Y_1) - E(Y_2) \\
 &= \int_0^1 \int_0^{y_1} y_1 \cdot 3y_1 \, dy_2 \, dy_1 - \int_0^1 \int_0^{y_1} y_2 \cdot 3y_1 \, dy_2 \, dy_1 \\
 &= \int_0^1 (3y_1^2) \, dy_1 - \int_0^1 \frac{3y_1^3}{2} \, dy_1 \\
 &= 3 \left[\frac{y_1^3}{3} \right]_0^1 - \frac{3}{2} \left[\frac{y_1^4}{4} \right]_0^1 \\
 &= \frac{3}{4} - \frac{3}{8} \\
 &= \underline{\underline{\frac{3}{8}}}
 \end{aligned}$$